AZURE CLOUD PROJECT — PROPOSAL SUBMISSION

Intelligent Serverless FinOps & Automated Compliance Remediation Orchestrator

Student Name: Mohammed Shamnan

Date: 31 August 2026

File Name: MOHAMMED_SHAMNAN_Azure_Project_Proposal.pdf

1. Project Title

Intelligent Serverless FinOps & Automated Compliance Remediation Orchestrator

2. Project Objective

• **What it aims to achieve:** Provide continuous automated auditing of cloud resource configurations against security benchmarks and cost-optimization standards, with instant auto-remediation.

• **What problem it addresses:** Cloud drift, misconfigured storage buckets, unencrypted disks, and orphaned resources that accumulate compliance fines and unnecessary costs.

• **Expected outcome:** An automated governance engine that detects cloud configuration drift within seconds and safely executes remediation workflows without human intervention.

3. Project Description

A serverless event-driven architecture that listens to Azure Resource Manager (ARM) activity logs via Event Grid. When a non-compliant resource is created or modified, an Azure Function evaluates the violation, triggers a remediation script (or approval workflow), and logs the audit event.

4. Azure Services

The project leverages the following 10 core Azure services to establish compute, data, integration, and security capabilities:

No.	Azure Service	Purpose in the Project
1	**Azure Event Grid**	Captures real-time Azure subscription management and resource state events.
2	**Azure Functions**	Executes event-driven compliance checking and auto-remediation code.
3	**Azure Policy**	Defines organizational compliance standards and security guardrails.
4	**Microsoft Defender for Cloud**	Continuously assesses security posture and scores resources.
5	**Azure Cosmos DB**	Maintains a historical ledger of compliance violations and remediation actions.
6	**Azure Logic Apps**	Orchestrates multi-step human-in-the-loop approval requests via Microsoft Teams or email.
7	**Azure Key Vault**	Stores webhook secrets, API keys, and service principal credentials securely.
8	**Azure Monitor**	Collects application metrics and operational health signals.
9	**Azure App Service**	Hosts the administrative management and reporting web portal.

No.	Azure Service	Purpose in the Project
10	**Azure Container Instances (ACI)**	Runs periodic batch security auditing tools and dependency vulnerability scanners.

5. Proposed Architecture

- **Network Components:** Secure virtual networks with private endpoints for Cosmos DB and Key Vault, outbound service endpoints for Event Grid, and IP restriction rules on App Service.
- **Data Flow:** Resource state change → Azure Activity Log → Event Grid → Azure Functions / Logic Apps → Cosmos DB audit store → Admin dashboard portal.

[Architecture Diagram Placeholder: Azure Activity Log feeding Event Grid, triggering Functions and Logic Apps for remediation and approval, writing audit records to Cosmos DB, surfaced via an admin dashboard portal.]

6. Implementation Plan

1. **Planning:** Establish compliance frameworks (e.g., CIS benchmarks, internal tagging policies) and define automated remediation actions.
2. **Azure Environment Setup:** Configure subscription-level Azure Policy definitions and diagnostic settings.
3. **Resource Deployment:** Deploy Event Grid topics, Function Apps, Cosmos DB databases, and Logic Apps workflows.
4. **Service Integration:** Connect Event Grid subscriptions to Function App triggers and integrate Logic Apps with Microsoft Teams connectors.
5. **Automation:** Deploy infrastructure using ARM/Bicep templates combined with continuous integration pipelines.
6. **Security:** Secure function execution using Managed Identities and store all sensitive tokens in Key Vault.
7. **Monitoring and Testing:** Intentionally deploy non-compliant resources (e.g., public storage accounts) to test end-to-end automated remediation speed.

7. Automation Plan

Selected Tool: ARM/Bicep templates and Azure PowerShell scripts.

Explanation: Automates deployment of Azure Policy definitions across management groups, auto-wiring of Event Grid subscriptions, and function code deployments.

8. Security & Monitoring Plan

- **Access Control & RBAC:** Contributor and Owner roles strictly limited; remediation functions assigned custom least-privilege RBAC roles capable of modifying only specific resource properties.
- **Network Security:** Outbound traffic from function apps restricted using Virtual Network Integration and Service Endpoints.
- **Security Configuration:** Code scanning integrated into CI/CD pipelines to prevent credential leakage.
- **Monitoring, Logging, & Alerts:** Function execution logs routed to Log Analytics; alerts configured for failed remediation attempts.

9. Expected Outcome

A fully operational compliance engine demonstrating real-time drift detection, instant auto-remediation of insecure resources, and comprehensive audit reporting dashboards.

10. Estimated Cost / Free Tier Usage

Designed to operate entirely within serverless consumption models (Consumption plan for Functions, serverless Cosmos DB), ensuring near-zero idle cost and full utilization of Azure free grant allotments.